

Demonstration of Proposed Features

Date	17 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
Maximum Marks	5 Marks

This section documents the live verification, functional workflows, and practical user scenarios demonstrated to confirm the capabilities of the Smart Lender AI application.

Demonstration Workflow Matrix

Workflow Phase	Details / Proven Operational Capabilities
Interactive Data Entry	Demonstrated smooth manual attribute collection (income ratios, credit profiles) via the responsive Bootstrap 5 web workspace interface.
Input Guard Evaluation	Proven client-side form interception and server-side filtering loops blocking negative numbers or extreme values instantly.
Live Real-Time Inference	Successful sub-second execution of the loaded pipeline serialization state (\$XGBoost\$/Random Forest parameters) yielding results under 140ms.
Dynamic Results Delivery	Visual validation of instant result updates via JavaScript Fetch requests, seamlessly outputting categorical badges and certainty percentages without full reloads.
Data Handling Privacy	Demonstrated complete in-memory data routing, showing that system runtime variables handle transient payloads securely without writing client traces to disk logs.

Feature Demonstration Checklist

S.No	Criteria / Core Verified Feature	Status
1	Web landing dashboard builds and loads cleanly across standard browsers	Yes
2		Yes

S.No	Criteria / Core Verified Feature	Status
	Frontend UI remains responsive and unblocked during heavy inference calls	
3	Invalid payload values trigger localized field warnings immediately	Yes
4	The final risk status badge clearly maps to the correct threshold bounds	Yes
5	Memory reclamation operates properly following rapid request inputs	Yes
6	Decoupled application logic handles model replacements without visual updates	Yes

Additional Notes / Comments

The application successfully passes all phase validation criteria. The execution runtime maintains sub-second responsiveness under concurrent operations, ensuring that prospective feature additions, such as multi-tenant role matrices or automated open-banking integrations, can deploy cleanly onto the current setup.